

1. Administrative & Study Identification

Full Study Title: Observational Secondary Data Study Describing Treatment With Dapagliflozin Among Adult Chronic Kidney Disease Patients **Short Title/Acronym:** OPTIMISE-CKD (OPTIMISE-CKD CEE)
Protocol Number: AstraZeneca Internal ID **NCT Number:** NCT06203704
Sponsor Name: AstraZeneca **Investigational Product:** Dapagliflozin (Forxiga) 10 mg **Phase of Development:** Observational / Phase IV
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize dapagliflozin utilization in clinical practice by describing treatment-naïve patients who are treated with dapagliflozin for CKD. **Secondary Objectives:** To describe selected outcomes of interest (e.g., CKD progression, cardio-renal events, hospitalizations) and treatment patterns among CKD patients treated with dapagliflozin up to 12 months post-initiation.

2.2 Background & Rationale It is important for physicians and patients to better understand dapagliflozin use in Chronic Kidney Disease (CKD) in routine clinical practice. Patients in real-world settings are often more heterogeneous and might have been underrepresented in interventional clinical trials. Additionally, payers require evidence from the broader CKD population prior to making reimbursement recommendations to enable access for all patients who have the approved CKD indication.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Multinational, longitudinal cohort study utilizing secondary data collection)
Control Type: Single-arm / Historical Control (Pre-post design)
Blinding: Open-label (Real-world data) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **NS:** 1,086 **Number of Sites:** Multinational (Europe and Central and Eastern Europe [CEE]) **Estimated Study Start:** Jan-2024 **Estimated Primary Completion:** Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (aged $\geq$ 18 years) at the time of dapagliflozin start. 2. Patient receives treatment

with dapagliflozin for CKD in accordance with the product label and local reimbursement criteria (CKD diagnosis based on KDIGO criteria: AER ≥ 30 mg/24 hours; ACR ≥ 30 mg/g; eGFR < 60 ml/min/1.73 m²). 3. Enrollment is to be performed a minimum of 30 days and a maximum of 90 days following the initiation of dapagliflozin 10 mg.

4.2 Exclusion Criteria 1. Diagnosed with type 1 diabetes or gestational diabetes mellitus at any time before the index date. 2. Prior treatment with dapagliflozin or another SGLT-2 inhibitor at any time before the index date. 3. Renal transplant, end-stage renal disease (eGFR < 15 ml/min/1.73 m²), or chronic dialysis on or at any time before the index date.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dapagliflozin 10 mg once daily **Route of Administration:** Oral **Duration of Treatment:** Up to 12 months post-initiation (Observation period)

5.2 Concomitant & Prohibited Therapy Permitted: Routine clinical treatments including RAASi (ACE inhibitors / ARBs) and other selected medications used for cardiovascular disease (CVD) and Type 2 Diabetes. **Prohibited:** Enrollment in a clinical trial that includes specific treatments as an investigational medicinal product on or 3 months before the index date.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Index	Day -90 to Day 0	Initiation of dapagliflozin. Extraction of baseline demographics, clinical characteristics, and laboratory measures from medical records.
Baseline	Day 30 to 90	Enrollment into the study cohort.
Interim Follow-up	Routine Care	Secondary data extraction of treatment patterns (dosage changes, RAASi add-ons) and healthcare resource utilization.
End of Study	Month 12	Final data extraction for cardio-renal outcomes, hospitalizations, and CKD progression evaluation.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Characterization of dapagliflozin utilization, evaluating baseline demographics, clinical characteristics, laboratory measures, and concomitant medication use. **Measurement Method:** Secondary data extraction from electronic or paper medical records.

7.2 Secondary & Exploratory Endpoints **Endpoint 1:** Treatment patterns of dapagliflozin, RAASi, and other selected medications,

including time to discontinuation, medication additions, and dosage changes. **Endpoint 2:** Selected clinical outcomes including CKD progression (advancement in stage), cardio-renal events (e.g., HF, nonfatal stroke, nonfatal MI), and healthcare resource utilization (all-cause and cardio-renal hospitalizations).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted from routine electronic health records and paper charts as part of standard clinical practice. **Serious Adverse Events (SAEs):** Handled in accordance with standard real-world pharmacovigilance practices. **Data Monitoring Committee (DMC):** N/A for secondary data observational studies.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible adult CKD patients (Intent-to-Observe cohort) newly initiated on dapagliflozin 10 mg identified in the secondary data sources. **Sample Size Justification:** $N = 1,086$. Designed to capture a broad, representative, heterogeneous population to establish real-world generalizability. **Handling of Missing Data:** Missing data will be quantified for all study variables, but there will be no attempts to impute them. Time to discontinuation will be estimated using the Kaplan-Meier method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoevidence Practice (GPP) and all applicable local regulations regarding secondary data collection. **Monitoring Plan:** Source data verification mapped to secondary database quality standards (electronic medical records). **Audit Trail:** Standard auditing of data extraction processes and database locking procedures.

11. Study Identifiers & Governance

Primary ID: OPTIMISE-CKD **Registry Number:** NCT06203704 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** OPTIMISE-CKD **Official Regulatory Title:** Observational Secondary Data Study Describing Treatment With Dapagliflozin Among Adult Chronic Kidney Disease Patients

12. Clinical Framework (CKD Specific)

Primary Condition: Chronic Kidney Disease (CKD) **Diagnosis**

Threshold: KDIGO criteria: AER ≥ 30 mg/24 hours; ACR ≥ 30 mg/g; eGFR < 60 ml/min/1.73 m²

Medical Methodology: Routine Clinical Care / Real-World Evidence **Optimization Target:**

Dapagliflozin 10 mg + standard background therapies (e.g., RAASi)

13. Study Procedures & Methodology

Management Pathway: Standard routine clinical pathway for CKD

Education Protocol (HCP): N/A (Observational tracking of natural prescription habits) **Education Protocol (Patient):** N/A (Standard of care self-management)

Quality Audit Mechanism: Electronic/paper medical record data extraction and review

14. Observation & Follow-Up Schedule

Total Duration: 12 Follow-up months **Onsite Visit Cadence:**

Dictated by routine clinical practice **Remote Monitoring**

Frequency: N/A (Retrospective/Prospective medical record review)

Checklist Review Interval: Data collected from index date up to 12 months post-initiation

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				